Power Is the Rate of Energy Change

Synopsis: Power is another physics term with a specific meaning/equation. Power (uppercase P is power, whereas lowercase p is momentum!), P, in watts (W) is P = W/t, where W is the work done (energy!) in joules (J) and t is time in seconds (s).

Work done, W = Fd, where F is the force in newtons (N) and d is the distance in metres (m), so rewriting, P = (Fd)/t = F(d/t) = Fv, where v is now the velocity (d/t): P = Fv.

Here is how you should think about power: the intensity of a "workout." You are in a gym, ready to do some step-up exercise. You are "energized" with a power bar and prepared to expend this energy on a stepper machine. You plan to do 100 step-ups. The intensity depends on how fast you finish the 100 step-ups. Completing the exercise in 1 minute is way more intense than 30 minutes. It is just like a sprint, which feels harder than a stroll, covering the same distance. An intense exercise involves expending energy at a higher rate. The level of exertion is correlated with intensity/power. Doing 100 step-ups faster is harder than doing it slowly, and this requires/takes more power.

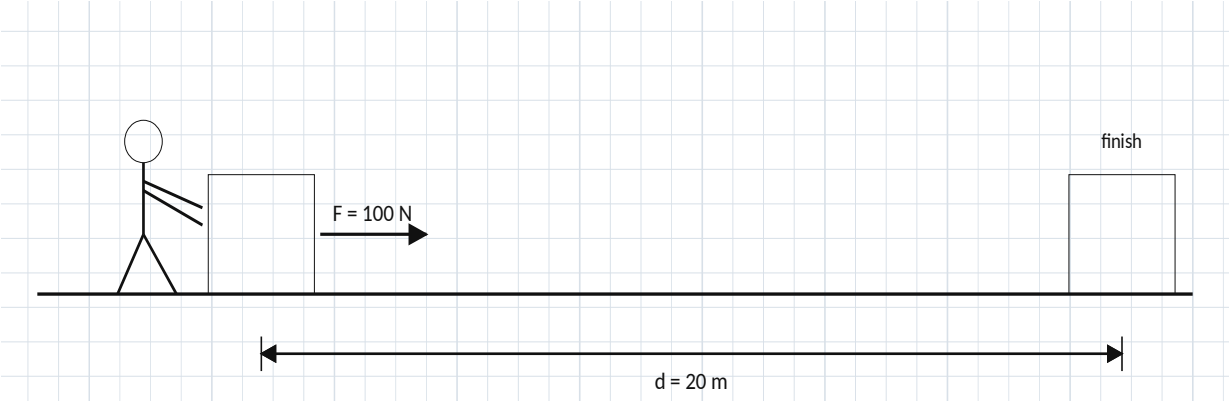
Performance Expectations: Calculate the rate of energy change, or power, when a force does work over a period of time. Use P = Fv for an object moving at steady speed. Convert between power and energy using time, and express the watt in fundamental units.

For PHY 151 — with calculus: Treat power as an instantaneous rate, P = dW/dt = F · v, and recover the energy from a power–time graph as the area under it.

Instructions and Questions

1) Power is how fast the work gets done.
   Work is the energy expended. Power says how fast you expended it. Push a box with 100 N for 20 m and you have done 2000 J of work whether it took you two minutes or ten seconds — the work does not care about the clock. The power does. High power means energy spent at a high rate: a faster workout, a sprint instead of a stroll.

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i) Calculate the work done by the pushing force over the 20 m.
ii) The push was finished in 120 s. Calculate the power.
iii) The same push is finished in 60 s instead. Calculate the power, and say what happened to it when the time halved.
iv) Now the same 2000 J spread over 1000 s. What sort of power is that? Give a number and a word.
v) In one sentence a classmate would follow: what does “high power” mean?

2) At steady speed, power is force times velocity.
P = W/t and W = Fd, so P = Fd/t. But d/t is the speed, so P = Fv. This is the form you want whenever something moves at a steady speed against a force — a car holding 25 m/s up a hill, an elevator rising, a boat against the drag of the water. The engine’s power is the force it supplies times the speed it holds.
i) A motor exerts 250 N of force to lift an object to a height of 5.6 m. If the time taken to do this is 4.3 s, how much power does the motor generate?
ii) A 1.2 kg solar car has a motor rated for 3.0 W. If the total oppositional force acting against the car when it is on a flat surface is 1.1 N, what speed will it travel?
iii) How much energy has that motor transferred after 20 s?
iv) The maximum speed a car can go up a hill is 25 m/s. If the power output of the engine is 150 kW, what is the total oppositional force acting against the car? Convert to watts first.
v) How much work does that engine do over 1.5 km of going up the hill?

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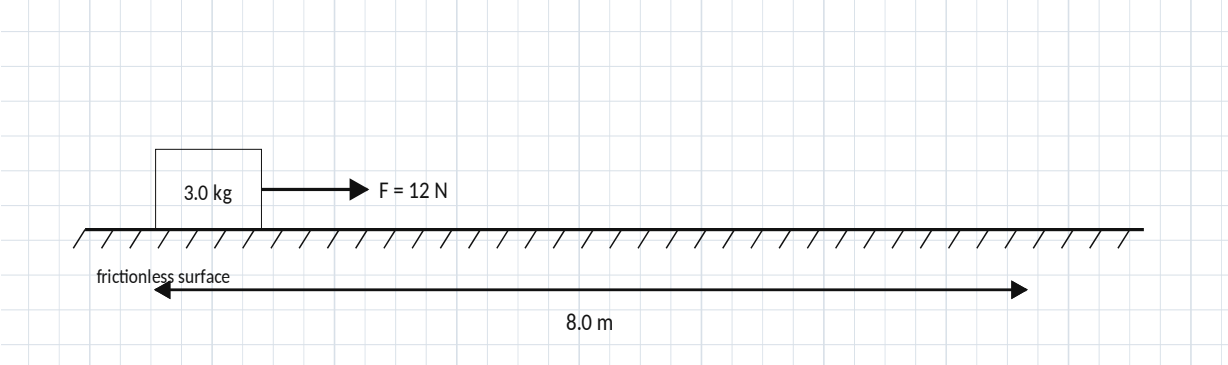
3) The watt, and getting energy back from power.
A watt is a joule per second. Turn the definition round and energy is power times time: $E = Pt$. A 60 W bulb spends 60 J every second, and a power station rated in megawatts is rated by how fast it delivers energy, not by how much. (The unit on your electricity bill, the kilowatt-hour, is a power times a time — it is an amount of energy: $1\ \text{kW} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J}$.)
   i) Express the unit of power (W) in terms of the fundamental units of time (s), distance (m), and mass (kg). [e.g. a newton (N) is equivalent to a $\text{kg}\cdot\text{m/s}^2$]
   ii) A power plant produces electrical power at a rate of 6.5 MW (megawatts). How much energy is transferred after 1 hour? (Convert time to seconds and power to watts.)
   iii) A 60 W bulb is left on for a whole day. How many joules is that? How many kilowatt-hours?

4) Who did more work, and who is more powerful?
These are two different questions, and the whole point of this Note is that they have different answers. Work asks how much energy changed hands. Power asks how fast. Keep them apart in each case below.
   i) Animal A and animal B both jump the same height, but B has 4 times the mass. However, animal A gets to that height in half the time. Which animal did more work? Which animal has more powerful legs? Explain.
   ii) Two elevators, A and B, move at constant speed. Elevator B moves with twice the speed of elevator A, and weighs twice as much. Compared to the power needed to lift elevator A, the power needed to lift elevator B is: the same / twice as great / half as great / four times as great.
   iii) If the time required for a student to swim 500 metres is doubled, the power developed by the student will be: halved / doubled / quartered / quadrupled.
   iv) As the power of a machine is increased, the time required to move an object a fixed distance: decreases / increases / remains the same.

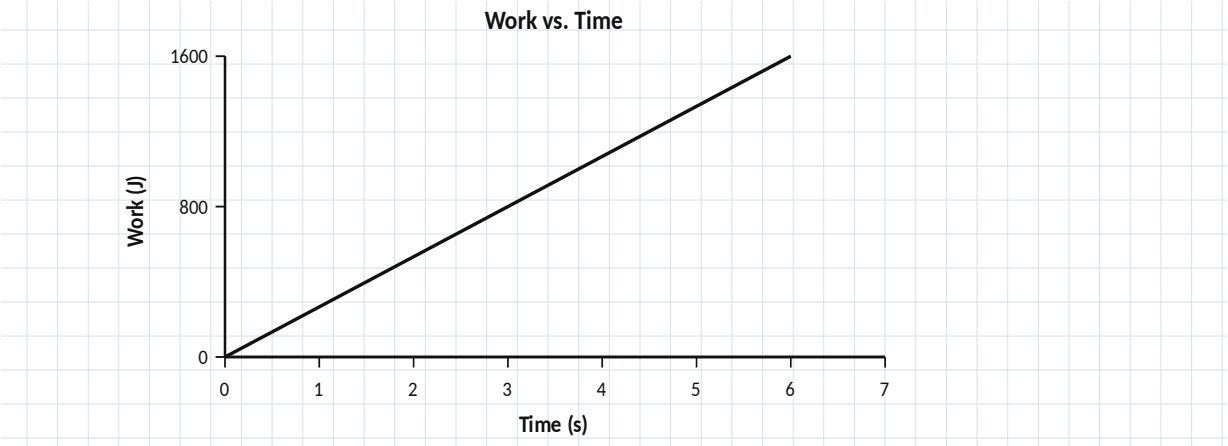
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v) The watt·second is a unit of: power / energy / impulse / work.

vi) A 5.8×10^4 W elevator motor can lift a total weight of 2.1×10^4 N with a maximum constant speed of how much?

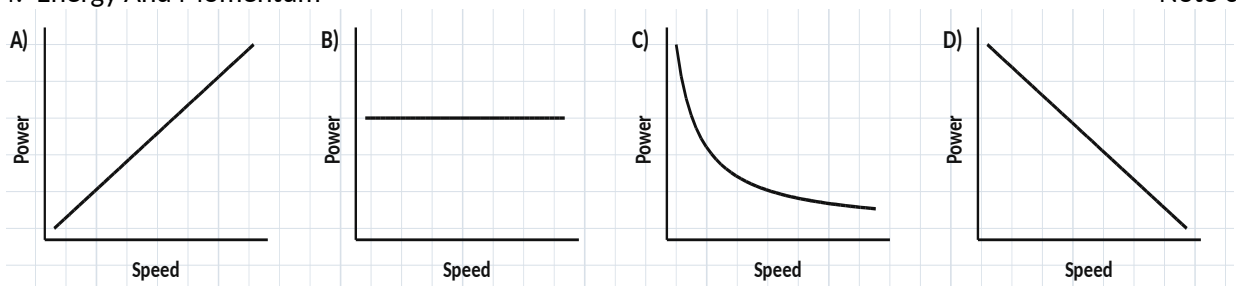


i) A 3.0 kg block is initially at rest on a frictionless, horizontal surface. The block is moved 8.0 m in 2.0 s by the application of a 12 N horizontal force, as shown. What is the average power developed while moving the block?



i) The graph represents the work done against gravity by a student as she walks up a flight of stairs at constant speed. How does the power output at time 2.0 s compare to that at time 4.0 s? Explain, or show a relevant calculation.

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i) Which graph best represents the relationship between the power required to raise an elevator and the speed at which the elevator rises?

Instantaneous power, and the area under a power–time graph.

P = W/t is an average over the whole job. The instantaneous version is the derivative, P = dW/dt, and since dW = F · dr the rate is P = F · v — a dot product, so only the part of the force along the velocity counts, exactly as only that part did work in Note 4. Run it backwards and the energy delivered between two times is the integral of the power, W = ∫ P dt: the area under a power–time graph.

i) A tow rope pulls a sledge at 3.0 m/s with a 200 N force at 30° above the horizontal. Calculate the power delivered to the sledge.
ii) An engine's output ramps up linearly from 0 to 40 kW over 8.0 s. Sketch P against t and find the energy delivered.
iii) The stairs graph in section 4 is a work–time graph. Say what its slope is, and read the power off it.

5) Looking ahead.

That closes the energy half of the module: work moves energy, conservation keeps the books, conservative forces earn a potential energy, and power says how fast it all happens. Next the module turns to the other conserved quantity we promised in Note 3 — momentum — and the preamble starts, as energy's did, with a quantity that is kept.

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